MobilityAPI

An OGC API – Moving Features server over MobilityDB

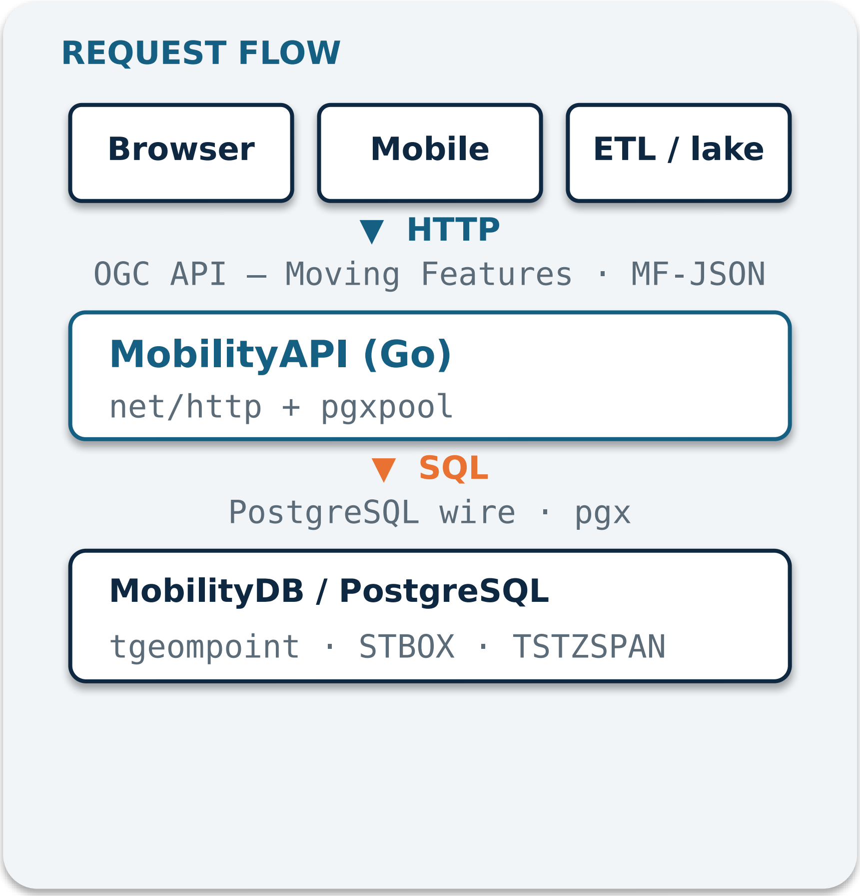
A thin compiled tier over MobilityDB · canonical Danish AIS example

`net/http + pgxpool · single static binary · no MEOS in the tier`

WHAT IT IS

A Thin Tier over MobilityDB, Fat Database

- **OGC-compliant:** implements OGC API – Moving Features – Part 1: Core (collections, items, temporal geometry, derived queries, temporal properties)
- **Thin compiled tier:** net/http and a pgx connection pool; a single static binary, no cgo and no MEOS in the process
- **Fat database:** every temporal computation stays in MobilityDB; the tier never parses or transforms a trajectory in memory
- **REST for any client:** browsers, mobile, microservices and ETL consume mobility data without SQL or the PostgreSQL wire protocol
- **Bounded as data grows:** streaming responses, keyset pagination and index-using filters keep memory and paging bounded as collections grow



Where the Work Happens

In the tier (Go)

- **Routing:** net/http with the Go 1.22 pattern mux; one handler per OGC resource
- **Pooling:** a pgx pool multiplexes requests onto a bounded set of database connections
- **Adapter:** crs URN to EPSG and interpolation Stepwise to Step, roughly fifty lines
- **Streaming:** responses are written row by row from a server-side cursor

In the database (MobilityDB)

- **Reads:** asMFJSON assembles MovingFeaturesJSON in SQL; the tier streams the text
- **Writes:** tgeompointFromMFJSON parses the payload into a tgeompoint
- **Filters:** bbox and datetime push to the GiST index; atTime clips
- **Measures:** speed, cumulativeLength and azimuth compute the derived properties

NO MEOS IN THE TIER

- The trajectory is never materialised or parsed in the tier
- It travels as MF-JSON text on the way out, MovingFeaturesJSON on the way in
- MobilityDB does the parsing, the geometry algebra and the temporal computations
- A single static binary, no cgo dependency

SAME DATA, TWO INTERFACES

MobilityDB SQL vs MobilityAPI REST

PostgreSQL / MobilityDB

```sql
INSERT INTO ships (id, mmsi, name, trip)
VALUES (205106000, 205106000, 'MARCUS',
  tgeompoint '[
    Point(645829 6058110)@2026-02-26 06:00+00,
    Point(651280 6058390)@2026-02-26 06:25+00,
    Point(656731 6058670)@2026-02-26 06:50+00]'
);
```

MobilityAPI

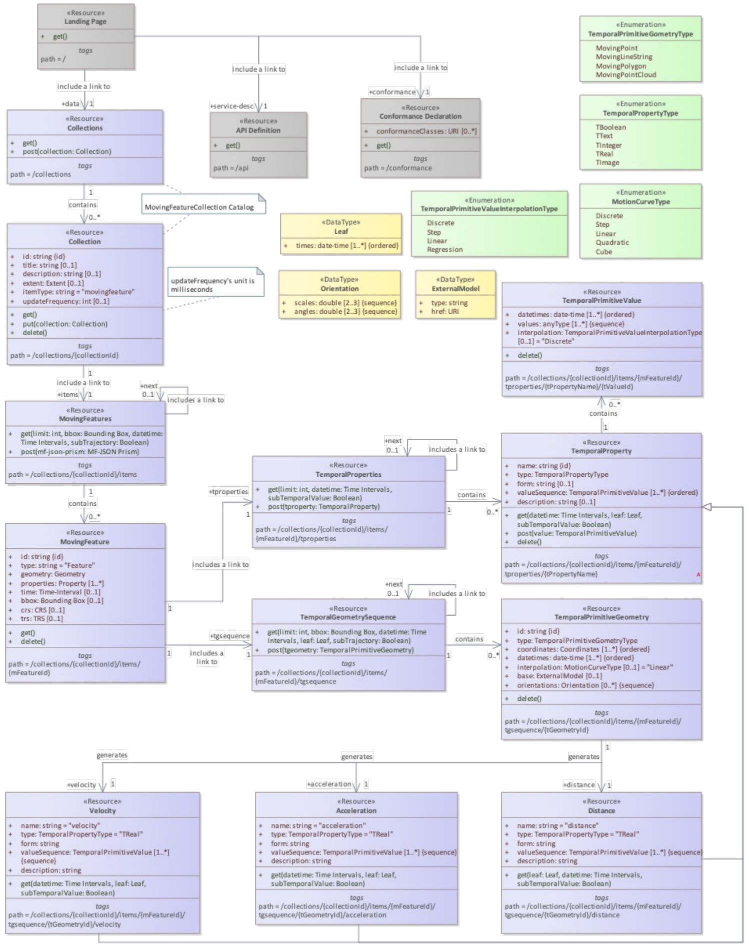
```
POST /collections/ships/items
Content-Type: application/json

{
  "type": "Feature",
  "id": "205106000",
  "properties": { "name": "MARCUS" },
  "temporalGeometry": {
    "type": "MovingPoint",
    "datetimes": ["2026-02-26T06:00:00+00", ...],
    "coordinates": [[645829, 6058110], ...],
    "interpolation": "Linear"
  }
}
```

Identical trajectory. The REST client never writes SQL or opens a database connection.

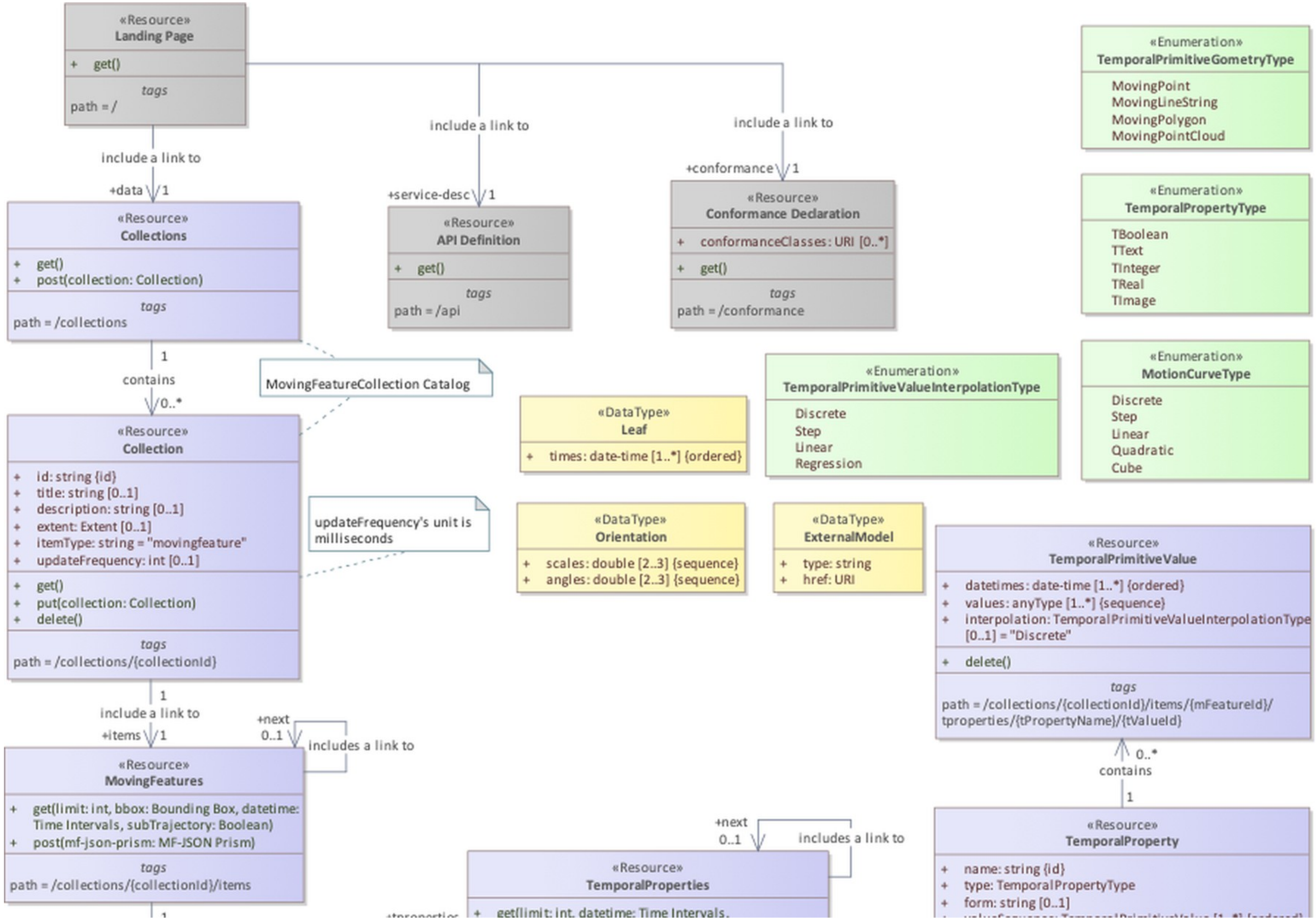
STANDARD

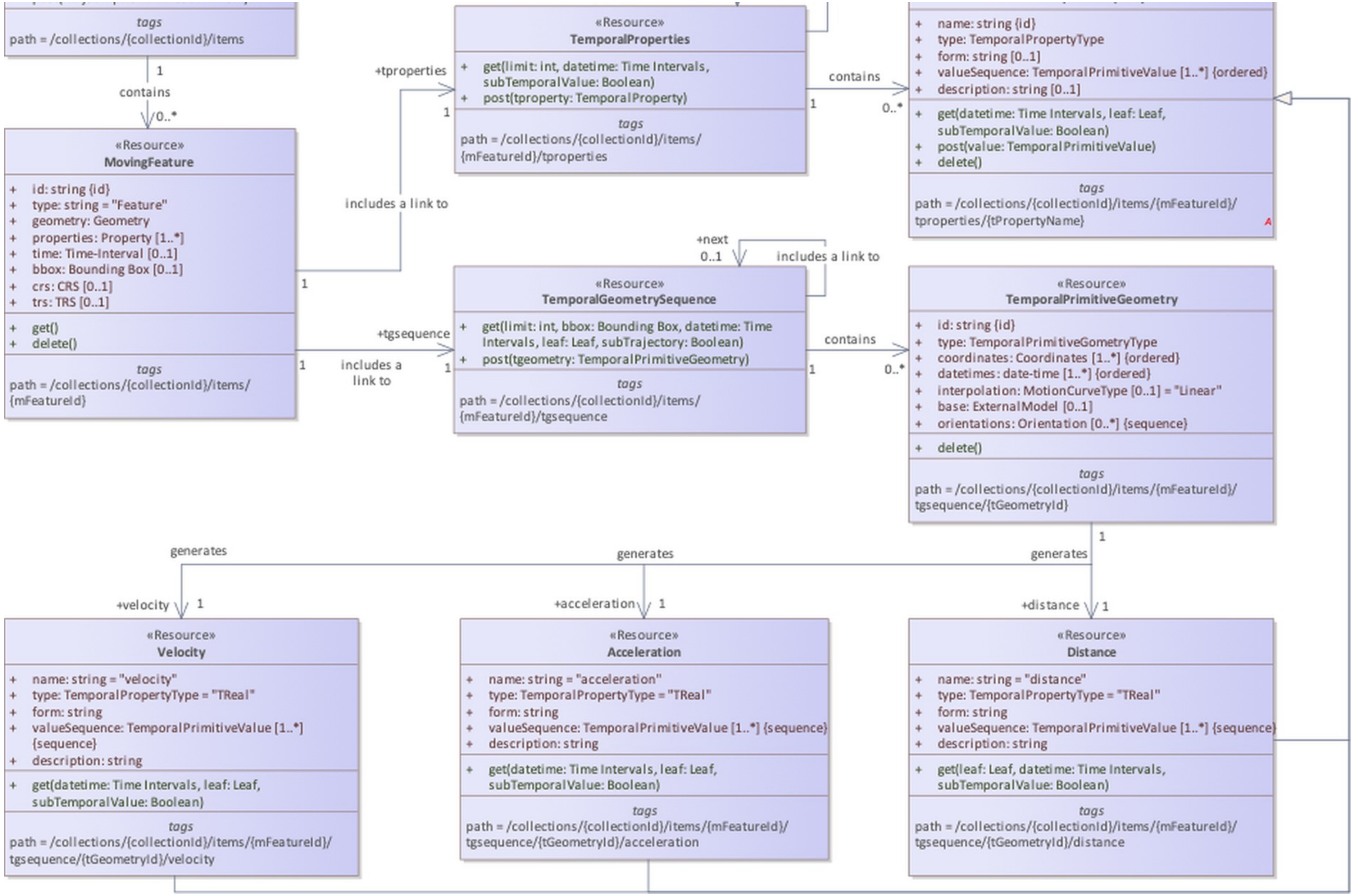
OGC API – Moving Features Data Model



Class diagram — OGC API – Moving Features – Part 1: Core (fig. 1)

- **Collection:** a named group of moving features (itemType movingfeature)
- **MovingFeature:** an item with id, properties, crs and trs
- **TemporalGeometry:** a MovingPoint with parallel coordinates and datetimes plus interpolation
- **Derived queries:** distance and velocity over a temporal geometry
- **TemporalProperties:** time-varying attributes derived from the trajectory


«Resource»
Landing Page
+ get()
tags
path = /
include a link to
+data 1
include a link to
+service-desc 1
include a link to
+conformance 1
«Enumeration»
TemporalPrimitiveGometryType
MovingPoint
MovingLineString
MovingPolygon
MovingPointCloud
«Resource»
Collections
+ get()
+ post(collection: Collection)
tags
path = /collections
«Resource»
API Definition
+ get()
tags
path = /api
«Resource»
Conformance Declaration
+ conformanceClasses: URI [0..*]
+ get()
tags
path = /conformance
«Enumeration»
TemporalPropertyType
TBoolean
TText
TInteger
TReal
TImage
1
contains
0..*
MovingFeatureCollection Catalog
«Enumeration»
TemporalPrimitiveValueInterpolationType
Discrete
Step
Linear
Regression
«Enumeration»
MotionCurveType
Discrete
Step
Linear
Quadratic
Cube
«DataType»
Leaf
+ times: date-time [1..*] {ordered}
«Resource»
Collection
+ id: string {id}
+ title: string [0..1]
+ description: string [0..1]
+ extent: Extent [0..1]
+ itemType: string = "movingfeature"
+ updateFrequency: int [0..1]
+ get()
+ put(collection: Collection)
+ delete()
tags
path = /collections/{collectionId}
updateFrequency's unit is milliseconds
«DataType»
Orientation
+ scales: double [2..3] {sequence}
+ angles: double [2..3] {sequence}
«DataType»
ExternalModel
+ type: string
+ href: URI
«Resource»
TemporalPrimitiveValue
+ datetimes: date-time [1..*] {ordered}
+ values: anyType [1..*] {sequence}
+ interpolation: TemporalPrimitiveValueInterpolationType [0..1] = "Discrete"
+ delete()
tags
path = /collections/{collectionId}/items/{mFeatureId}/tproperties/{tPropertyName}/{tValueId}
1
include a link to
+items 1
+next
0..1
includes a link to
0..*
contains
1
«Resource»
MovingFeatures
+ get(limit: int, bbox: Bounding Box, datetime: Time Intervals, subTrajectory: Boolean)
+ post(mf-json-prism: MF-JSON Prism)
tags
path = /collections/{collectionId}/items
+next
0..1
includes a link to
«Resource»
TemporalProperties
«Resource»
TemporalProperty
+ name: string {id}
+ type: TemporalPropertyType
+ form: string [0..1]


tags
path = /collections/{collectionId}/items
1
contains
0..*
«Resource»
MovingFeature
+ id: string {id}
+ type: string = "Feature"
+ geometry: Geometry
+ properties: Property [1..*]
+ time: Time-Interval [0..1]
+ bbox: Bounding Box [0..1]
+ crs: CRS [0..1]
+ trs: TRS [0..1]
+ get()
+ delete()
tags
path = /collections/{collectionId}/items/{mFeatureId}
includes a link to
+tproperties
1
1
«Resource»
TemporalProperties
+ get(limit: int, datetime: Time Intervals, subTemporalValue: Boolean)
+ post(tproperty: TemporalProperty)
tags
path = /collections/{collectionId}/items/{mFeatureId}/tproperties
contains
1
0..*
+ name: string {id}
+ type: TemporalPropertyType
+ form: string [0..1]
+ valueSequence: TemporalPrimitiveValue [1..*] {ordered}
+ description: string [0..1]
+ get(datetime: Time Intervals, leaf: Leaf, subTemporalValue: Boolean)
+ post(value: TemporalPrimitiveValue)
+ delete()
tags
path = /collections/{collectionId}/items/{mFeatureId}/tproperties/{tPropertyName}
+next
0..1
includes a link to
+tgsequence
1
includes a link to
1
«Resource»
TemporalGeometrySequence
+ get(limit: int, bbox: Bounding Box, datetime: Time Intervals, leaf: Leaf, subTrajectory: Boolean)
+ post(tgeometry: TemporalPrimitiveGeometry)
tags
path = /collections/{collectionId}/items/{mFeatureId}/tgsequence
contains
1
0..*
«Resource»
TemporalPrimitiveGeometry
+ id: string {id}
+ type: TemporalPrimitiveGeometryType
+ coordinates: Coordinates [1..*] {ordered}
+ datetimes: date-time [1..*] {ordered}
+ interpolation: MotionCurveType [0..1] = "Linear"
+ base: ExternalModel [0..1]
+ orientations: Orientation [0..*] {sequence}
+ delete()
tags
path = /collections/{collectionId}/items/{mFeatureId}/tgsequence/{tGeometryId}
1
generates
generates
generates
+velocity
1
+acceleration
1
+distance
1
«Resource»
Velocity
+ name: string = "velocity"
+ type: TemporalPropertyType = "TReal"
+ form: string
+ valueSequence: TemporalPrimitiveValue [1..*] {sequence}
+ description: string
+ get(datetime: Time Intervals, leaf: Leaf, subTemporalValue: Boolean)
tags
path = /collections/{collectionId}/items/{mFeatureId}/tgsequence/{tGeometryId}/velocity
«Resource»
Acceleration
+ name: string = "acceleration"
+ type: TemporalPropertyType = "TReal"
+ form: string
+ valueSequence: TemporalPrimitiveValue [1..*] {sequence}
+ description: string
+ get(datetime: Time Intervals, leaf: Leaf, subTemporalValue: Boolean)
tags
path = /collections/{collectionId}/items/{mFeatureId}/tgsequence/{tGeometryId}/acceleration
«Resource»
Distance
+ name: string = "distance"
+ type: TemporalPropertyType = "TReal"
+ form: string
+ valueSequence: TemporalPrimitiveValue [1..*] {sequence}
+ description: string
+ get(leaf: Leaf, datetime: Time Intervals, subTemporalValue: Boolean)
tags
path = /collections/{collectionId}/items/{mFeatureId}/tgsequence/{tGeometryId}/distance

How MobilityAPI Persists Moving Features

STORED COLUMNS

collections

PK **id**	TEXT
title	TEXT
description	TEXT
item_type	TEXT
crs	INTEGER

ships

PK **id**	INTEGER
mmsi	BIGINT
name	TEXT
trip	tgeompoint

one feature table per collection

STANDARD FIELDS SERVED

Collection

- **Stored:** id, title, description, itemType, crs
- **Derived:** extent (spatial bbox and time interval)
- **Generated:** links

MovingFeature

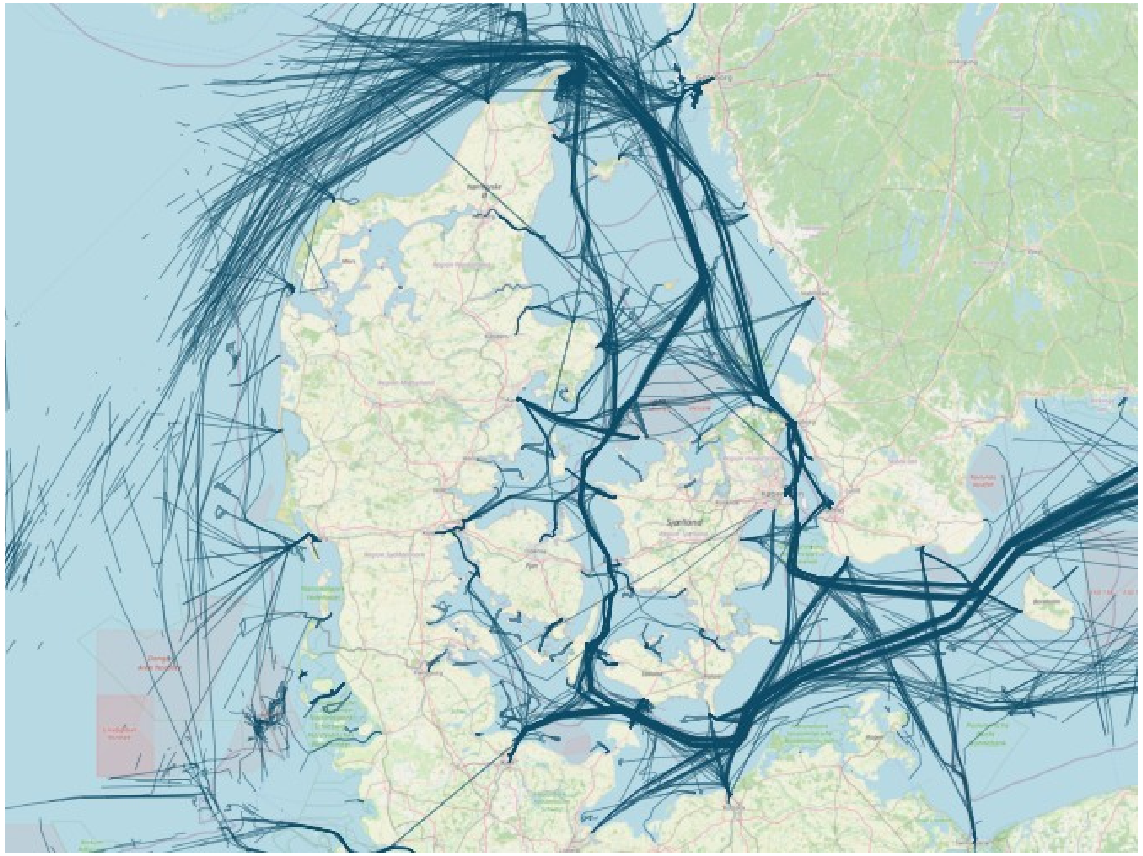
- **Stored:** id, properties (mmsi, name)
- **Derived from trip:** temporalGeometry, bbox, time, geometry
- **From the registry:** crs
- **Constant:** trs
- **Generated:** links

Derived fields come O(1) from the tgeompoint and its inline STBOX; links are generated per request.

CANONICAL EXAMPLE

A Day of Danish AIS Ship Traffic

- **Source:** Danish Maritime Authority AIS feed (aisdata.ais.dk)
- **Window:** 2026-02-26 to 2026-02-27
- **2,833 trajectories:** from 1,878 distinct vessels (MMSI), each a MovingPoint
- **6.2 million instants:** of observed position across the collection
- **CRS:** EPSG:25832 (ETRS89 / UTM 32N, metres)



Danish AIS vessel trajectories, 2026-02-26

REFERENCE

The OGC Endpoint Surface

GET POST `/collections`

GET `/collections/{collectionId}`

GET POST `/collections/{collectionId}/items`

GET DELETE `/collections/{collectionId}/items/{mFeatureId}`

GET POST `/collections/{collectionId}/items/{mFeatureId}/tgsequence`

GET `/collections/{collectionId}/items/{mFeatureId}/tgsequence/{tGeometryId}/distance`

GET `/collections/{collectionId}/items/{mFeatureId}/tgsequence/{tGeometryId}/velocity`

GET `/collections/{collectionId}/items/{mFeatureId}/tproperties`

GET `/collections/{collectionId}/items/{mFeatureId}/tproperties/{tPropertyName}`

GET `/` · `/api` · `/conformance`

Every path above is OGC API – Moving Features – Part 1: Core. acceleration is offered at the same path but returns 501 (not derivable for this motion model). Items query parameters: bbox, datetime, subtrajectory, leaf, limit, page cursor.

WALKTHROUGH

The Ships Collection, End to End

List → filter → retrieve → derive → write, all over HTTP

List Collections · Retrieve One

GET `/collections`

200 OK


```json
{
  "collections": [
    {
      "id": "ships",
      "title": "ships",
      "itemType": "movingfeature",
      "description": "Danish AIS vessel trajectories",
      "links": [
        { "rel": "items",
          "href": "/collections/ships/items" },
        { "rel": "enclosure",
          "href": "/collections/ships/export",
          "type": "application/x-ndjson" }
      ]
    }
  ]
}
```


GET `/collections/ships`

200 OK


```json
{
  "id": "ships", "title": "ships",
  "itemType": "movingfeature",
  "description": "Danish AIS vessel trajectories",
  "crs": [".../def/crs/EPSG/0/25832"],
  "extent": {
    "spatial":  { "bbox": [[152625, 5942006,
                            1020003, 6545239]] },
    "temporal": { "interval": [["2026-02-26 ...",
                               "2026-02-27 ..."]] } },
  "links": [ { "rel": "self" },
             { "rel": "items" },
             { "rel": "enclosure" } ]
}
```


- extent is the spatial bbox and time interval of the collection
- crs and links round out the standard collection resource

Stream the FeatureCollection, Page by Keyset

GET `/collections/ships/items?limit=2`

```
200 OK   Content-Type: application/json
{
  "type": "FeatureCollection",
  "features": [
    { "id": "1", "type": "Feature",
      "properties": { "mmsi": 111219510, "name": "vessel 111219510" },
      "crs": { "type": "Name", "properties": {
        "name": "urn:ogc:def:crs:EPSG::25832" } },
      "bbox": [550889, 6328752, 616716, 6346258],
      "time": ["2026-02-26 13:28:08+01", "2026-02-26 14:41:04+01"],
      "temporalGeometry": { "type": "MovingPoint",
        "sequences": [ { "datetimes": [...], "coordinates": [...] } ],
        "interpolation": "Linear" },
      "links": [ { "rel": "self", "href": ".../items/1" } ] },
    { "id": "2", "type": "Feature", ... }
  ],
  "numberReturned": 2,
  "links": [ { "rel": "next",
               "href": "/collections/ships/items?after=2&limit=2" } ]
}
```

- **Streamed:** each Feature is written from a server-side cursor; memory is bounded for any page size
- **Keyset next:** the next link advances by id (after=), with no OFFSET
- **asMFJSON:** the temporalGeometry is assembled in SQL and streamed as text
- **limit:** caps the page; the tier enforces a maximum

Retrieve a Single Vessel

GET `/collections/ships/items/1`


```
200 OK
{
  "id": "1", "type": "Feature",
  "properties": { "mmsi": 111219510, "name": "vessel 111219510" },
  "crs": { "type": "Name", "properties": {
    "name": "urn:ogc:def:crs:EPSG::25832" } },
  "trs": { "type": "Link",
           "properties": { "type": "ogcdef",
             "href": ".../uom/ISO-8601/0/Gregorian" } },
  "bbox": [550889, 6328752, 616716, 6346258],
  "time": ["2026-02-26 13:28:08+01", "2026-02-26 14:41:04+01"],
  "geometry": { "type": "MultiLineString", "coordinates": [...] },
  "temporalGeometry": {
    "type": "MovingPoint",
    "sequences": [ { "datetimes": [...], "coordinates": [...] } ],
    "interpolation": "Linear" },
  "links": [ { "rel": "self",
               "href": "/collections/ships/items/1" } ]
}
```


- **One movement:** the full trajectory of one vessel as MovingFeaturesJSON
- **crs:** carried as an OGC URN; the tier maps to and from the EPSG form
- **interpolation:** Linear, so the vessel moves continuously between fixes
- **sequences:** one per continuous run; gaps split the trajectory

SPATIAL FILTER

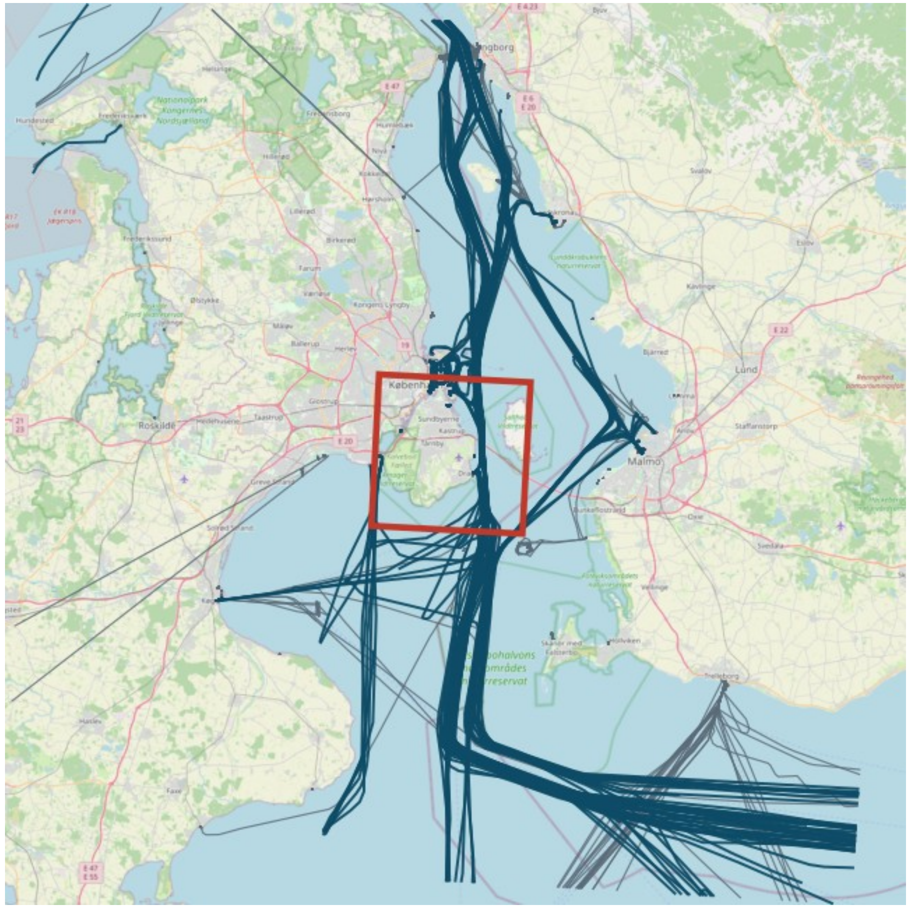
Vessels Intersecting a Bounding Box

GET `/collections/ships/items?bbox=720000,6160000,738000,6178000`

```
GET /collections/ships/items
      ?bbox=720000,6160000,738000,6178000
      &limit=10000

// bbox = minx,miny,maxx,maxy in the storage CRS
// EPSG:25832 (metres)
// 170 of 2,833 trajectories cross the box
```

- **bbox:** keeps vessels whose trajectory intersects the envelope
- **Index:** pushed to the MobilityDB GiST index via the overlap operator
- **CRS:** coordinates are in the collection's storage CRS (EPSG:25832)
- **Composable:** with datetime, subtrajectory, limit and after



Trajectories crossing the box (teal) over other traffic (grey); the box in red

TEMPORAL SLICE

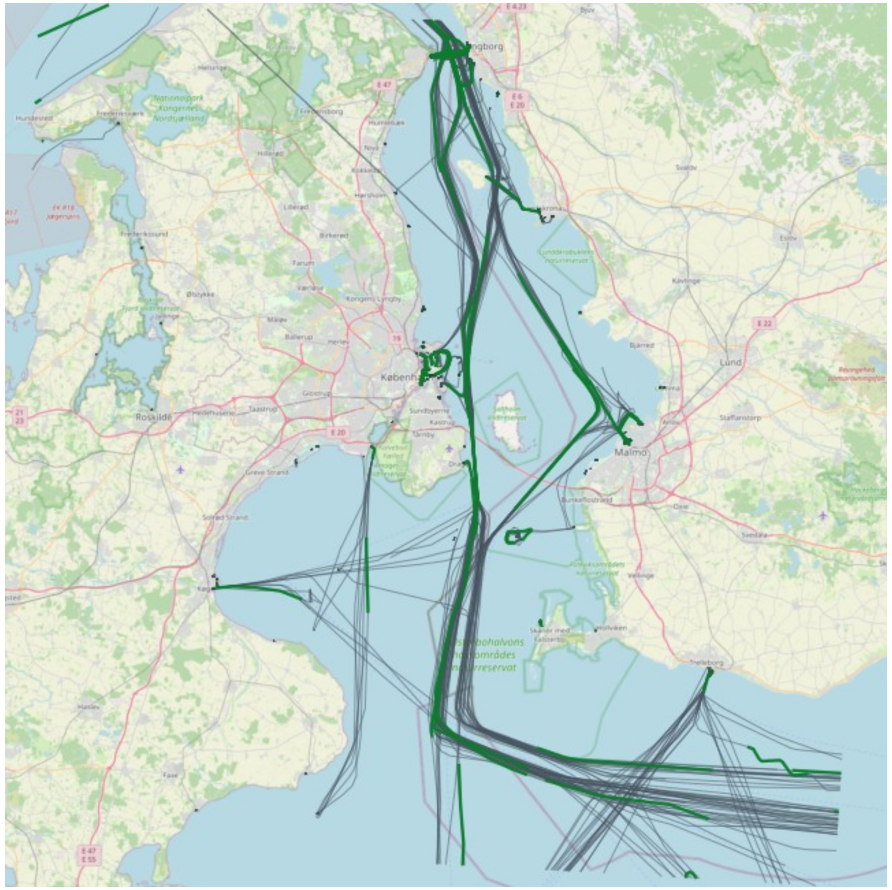
Restrict Trajectories to a Time Interval

GET `/collections/ships/items?subtrajectory=true`

```
GET /collections/ships/items
      ?subtrajectory=true
      &datetime=2026-02-26T14:00:00+01
               /2026-02-26T15:00:00+01

// datetime is an RFC 3339 interval (start/end)
// subtrajectory=true clips each trajectory to the
// window with atTime, instead of filtering trips
```

- **subtrajectory:** clips each trajectory to the datetime interval via atTime
- **datetime:** RFC 3339 interval; a bare instant is also accepted
- **Gap aware:** rows whose values fall entirely in a temporal gap are dropped
- **Validation:** subtrajectory requires a bounded interval



Subtrajectories within a one-hour window (green) over full trajectories (grey)

WRITES

Create · Append · Delete

POST `/collections/ships/items`

```
{ "type": "Feature", "id": "900001",
  "properties": { "name": "demo" },
  "temporalGeometry": {
    "type": "MovingPoint", "interpolation": "Linear",
    "datetimes": ["2026-02-26T10:00:00+01", ...],
    "coordinates": [[600000, 6330000], ...] } }

201  { "message": "created", "id": "900001" }
```

POST `/collections/ships/items/900001/tgsequence`

```
// append a temporally-disjoint sub-trajectory
200  { "message": "appended", "id": "900001" }

// overlapping the existing time span:
409  { "code": "409",
       "description": "the sub-trajectory overlaps
        the existing temporal geometry in time" }
```

DELETE `/collections/ships/items/900001`

204 remove the feature

- **merge:** append uses MobilityDB merge, which rejects time overlap; the tier maps that to 409
- **Derived properties:** are computed from the geometry, so writes to them return 501; modify the geometry instead
- **Single geometry:** a feature carries one temporal geometry; delete the feature rather than a primitive

The Temporal-Geometry Sequence

GET `/collections/ships/items/1/tgsequence`

```
200 OK
{
  "type": "MovingPoint",
  "crs": { "type": "Name", "properties": {
    "name": "urn:ogc:def:crs:EPSG::25832" } },
  "sequences": [
    { "datetimes": ["2026-02-26T13:28:08+01", ...],
      "coordinates": [[721000.4, 6178001.2], ...],
      "lower_inc": true, "upper_inc": true }
  ],
  "interpolation": "Linear"
}
```

Selectors on this resource

- **datetime:** clip the sequence to an interval (atTime)
- **leaf:** return the geometry at one or more instants
- **POST:** append a temporally-disjoint sub-trajectory

The temporal geometry is the trajectory itself, returned as MovingFeaturesJSON straight from asMFJSON.

Distance · Velocity · Acceleration

GET `/collections/ships/items/1/tgsequence/0/velocity`


```
200 OK
{ "name": "velocity", "type": "TReal", "form": "m/s",
  "description": "Speed over ground ...",
  "valueSequence": [
    { "datetimes": ["2026-02-26T13:28:08+01", ...],
      "values": [22.78, 11.03, ...],
      "interpolation": "Stepwise" } ],
  "links": [ { "rel": "self", ... } ] }
```


GET `/collections/ships/items/1/tgsequence/0/acceleration`


```
501
{ "code": "501", "description":
  "acceleration is not derivable: linearly
   interpolated position gives a piecewise-constant
   speed, whose derivative is zero within each
   segment and undefined at the vertices" }
```


- **velocity:** speed over ground from speed(trip), piecewise-constant (Stepwise)
- **distance:** cumulative length from cumulativeLength(trip) (Linear)
- **acceleration:** not fabricated, because with linear position the speed is a step function whose derivative is not a finite function
- **Exact:** each measure is a MobilityDB function reshaped into an OGC temporalProperty (TReal)

Time-Varying Attributes per Feature

GET `/collections/ships/items/1/tproperties`

```
200 OK
{
  "temporalProperties": [
    { "name": "velocity", "type": "TReal", "form":
"m/s" },
    { "name": "distance", "type": "TReal", "form": "m" },
    { "name": "heading",  "type": "TReal", "form":
"rad" }
  ],
  "numberReturned": 3,
  "links": [ { "rel": "self", ... } ]
}
```

GET `/collections/ships/items/1/tproperties/velocity?leaf=...`

```
{ "name": "velocity", "type": "TReal", "form": "m/s",
  "valueSequence": [
    { "datetimes": ["2026-02-26T13:28:08+01"],
      "values": [22.78],
      "interpolation": "Discrete" } ] }
```

Property type

TBoolean · TText · TInteger · TReal · TImage

- **Derived:** velocity, distance and heading are computed from the trajectory, read-only
- **valueSequence:** datetimes and values with each segment's true interpolation
- **leaf:** selects values at given instants; the result is Discrete
- **datetime:** clips the property to an interval

Declared Classes, API Definition, Errors

Conformance classes

- movingfeatures / conf / common
- / conf / mf-collection
- / conf / movingfeatures
- ogcapi-common-1 / conf / core
- ogcapi-common-2 / conf / collections

API definition

- **/api:** an OpenAPI 3.0 definition
- **service-desc:** linked from the landing page
- **Landing:** self, service-desc, conformance, data
- Clients discover the surface at runtime

OGC error responses

- **400:** invalid parameters or body
- **404:** collection / feature not found
- **409:** sub-trajectory overlaps in time
- **501:** measure not derivable / derived write
- Body: { "code", "description" }

One Streaming Tier, Standard and Beyond

393 MB

in one streamed response

≈18 MB

resident, bounded by the stream not the result size

STANDARD ITEMS RESOURCE

- **Streaming:** the FeatureCollection is written row by row from a cursor, so memory is bounded for any size
- **Keyset pagination:** next links advance by id with no OFFSET, constant cost through large collections
- **Index-using filters:** bbox and datetime push to the MobilityDB GiST index; subtrajectory clips with atTime

EXTENSIONS — NOT IN conformsTo

GET `/collections/{collectionId}/export`

NDJSON, or Parquet with trajectory WKB + bbox/time sidecar; enclosure link

PUT `/collections/{collectionId}/items/{mFeatureId}`

in-place replace; the standard allows only GET and DELETE on a feature

Streaming keeps memory bounded for the standard responses and the lakehouse export; a step toward the Data Lake House architecture in the MobilityDB ecosystem.

FUTURE WORK

One Tier, Many MEOS Engines

Thin OGC API – Moving Features tier

MF-JSON / WKB wire format · streaming · keyset paging · index pushdown · no MEOS in the tier

▼ serves any MEOS-backed engine; the engine becomes a configuration choice

MobilityDB

`PostgreSQL`

Transactional serving over a live database

Implemented today

MobilityDuck

`DuckDB`

Embedded, Parquet-native lakehouse queries

Planned

MobilitySpark

`Apache Spark`

Distributed batch over very large histories

Planned

MobilityFlink

`Apache Flink`

Streaming, real-time moving features

Planned

MEOS types and the MF-JSON / WKB wire format are shared across the ecosystem, so the same tier fronts a transactional database today and the lakehouse and streaming engines as they mature.

RECAP

The Standard Surface, Recapped

COLLECTIONS

- ✓ list (GET)
- ✓ retrieve (GET)
- ✓ registry-backed

MOVING FEATURES

- ✓ insert (POST)
- ✓ retrieve / stream
- ✓ delete (DELETE)
- ✓ keyset paging

QUERYING

- ✓ bbox spatial filter
- ✓ datetime interval / instant
- ✓ subtrajectory clipping
- ✓ leaf selector

TEMPORAL GEOMETRY

- ✓ get sequence (GET)
- ✓ append sub-trajectory (POST)
- ✓ datetime / leaf clip

DERIVED MEASURES

- ✓ distance
- ✓ velocity
- ✓ acceleration → 501 (not derivable)
- ✓ exact, as TReal

CONFORMANCE & SCALE

- ✓ 5 conformance classes
- ✓ OpenAPI at /api
- ✓ streaming + keyset
- ✓ index-using filters

Thanks for Listening

Questions?

REFERENCES

Find Out More

- **OGC API – Moving Features – Part 1: Core**
  https://docs.ogc.org/is/22-003r3/22-003r3.html
- **MobilityDB**
  https://github.com/MobilityDB/MobilityDB · https://mobilitydb.com
- **MEOS / ecosystem overview**
  https://libmeos.org
- **AIS data — Danish Maritime Authority**
  http://aisdata.ais.dk